Name.....

Signature……..**Candidates Personal Number**…….

545/1

CHEMISTRY.

Paper 1.

Aug /2025.

2 hours

CHEMISTRY DEPARTMENT

New Lower Secondary Chemistry Examination

Uganda Certificate of Education

Internal mock set 6

Paper 1

2 hours

INSTRUCTIONS:

This paper consists of **two** *sections* **A** *and* **B**. *It has* **six** *examination items.*

Section **A** *has* **one compulsory** items.

Section **B** *has* **two parts, I** and **II**. *Answer* **one** item from each part.. Answer **four** items in all.

Answers to section **A must** *be written in the spaces provided while those of section* **B must** *be written on the provided answer sheets.*

Any additional item(s) answered will **not** *be scored*

Where necessary use, Mg = 24 , Cl = 35.5 , *1 mole of gas occupies* 22.4 dm3 at s.t.p

SECTION A

Answer **all** *items in this section in the spaces provided.*

Item 1

After a very busy day, Bushirah developed persistent aches characterized by muscle and back strain in the evening. Her mother advised her to use a product **X** ,an extract of mixture of leaves from two plant species, she herself had used previously when she got the same problem Unfortunately, it only brought her a brief bit of ease, wearing off after four hours. In the morning, she went to nearby healthy facility where she was prescribed a new product **Y** that provided definitive and effective resolution to her ailment.

Task. As a chemistry student,

(a) Categorize the products she used.

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(b) State how the new product enabled her to feel well.

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(c) advise her on that dangers associated with use of the first product.

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(d) evaluate the products she used .

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Item 2.

An investor intends to set up a chemical company, whose production processes will utilize heating . To optimize these processes, he must first understand the melting trend of a few elements and how to use these elements responsibly . Given the table below.

Period 3 elements	**Na**	**Mg**	**Al**
Melting points (°C)	88.2	1091	2470

For preliminary experiment ,small scale experiment they plan to produce 500g of magnesium chloride, a common dietary supplement reacting chlorine gas over heated magnesium as equation below. He is interested to know the volume of gas at s.t.p required for the reaction . You have been invited guidance

$$Mg(s) + Cl_2(g) \longrightarrow MgCl_2(s)$$

Task: As a Chemistry student,

(a) explain to him the category of elements in the table..

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(b) guide him to understand the trend in melting points of the given elements

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(c) Calculate the volume of gas required for the reaction.

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(d) advise him the environmental impact associated with use of higher melting point element.

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SECTION B

Part I (*Attempt one item from this part*)

Item 3. Soapy detergents are cleansing products in every home. Due to high demand of the detergents on market, the government has cleared an investor to set up a mega soapy detergent manufacturing plant in Masaka city. However ,the locals are concerned about its impact and are seeking assurance on how they will benefit from it's operations. The city officials have organized a workshop to educate the locals. You have been invited to attend.

Task. *As* a chemistry student, make a write up you will use during the presentation.

Item 4.

The government has cleared an investor to establish an oxygen gas production plant in Nakawa – Kampala .This move is intended to maintain continuous supply of oxygen gas needed by many hospitals for patients under anesthesia and having operation . However, residents are concerned about the potential for environmental damage from production process and are questioning how they will directly benefit from the plant's operations. The official from Uganda Investments Authority (UIA) have organized workshops to address the locals . You have been invited to address the locals .

Task. Write a write up presentation you will use during the presentation.

Part II

. **(*Attempt one item from this part.*)**

Item 5. The National Environmental Management Authority , (NEMA) officials have noted with concern about the heavy reliance on keresone and wood charcoal for cooking in many homes which has posed serious environmental consequences. This has prompted a coordinated effort to develop and implement a joint action plan aimed at educating the public and mitigating these effects. . They have organized radio talk shows to create awareness to the public. You have been invited to address the public.

Task. As a chemistry prepare a short presentation you will deliver in the radio talk show

Item 6 Many **igneous** rocky areas in different regions of Uganda have been over exploited by locals due to due to useful substances they contain used for different purposes.. This has raised significant alarm among environment officials who are worried about the potential consequences of these actions to water and air quality in such regions. The government has organized workshops to raise awareness amongst the locals living in such areas about their actions. . You have been invited to attend one of the workshops and address the locals.

Task. As a chemistry student, prepare a short message you will use to deliver in the workshop.

End.